Estimation of Coherence and Cross Spectra

Copyright © Barry Van Veen 2014

AllSignalProcessing.com

Key Concepts

1) The cross spectrum is similar to the power spectrum in that it is the discrete-time Fourier transform of the cross covariance.

2) Estimation of cross spectra is similar to estimation of power spectra.

3) Periodogram-inspired estimates of the cross spectrum suffer from high variance, just like the periodogram.

4) Coherence and magnitude-squared coherence (MSC) are defined in terms of the cross and power spectrum.

5) Periodogram inspired estimates of the MSC fail - they always give MSC of 1.

6) Averaging approaches are desirable for estimating cross spectra and mandatory for estimating MSC.

7) Welch's method for MSC

    a) Divide both signals into segments.

b) Take the DFT of each segment.

c) Use Welch's method to find the cross and power spectra by averaging products of DFTs over segments.

d) Compute MSC from estimated cross and power spectra.

8) Variance decreases as the number of segments increase.

9) The MSC is non negative, so high variance can lead to positive bias where the MSC should be small or zero.

Video Screenshots

Cross Spectrum

$$r_{xy}[k] = E\{x[n]\, y[n-k]\} \overset{\text{DTFT}}{\longleftrightarrow} S_{xy}(\omega)$$

Coherence

$$\text{Coh}_{xy}(\omega) = \frac{S_{xy}(\omega)}{S_{xx}^{1/2}(\omega)\, S_{yy}^{1/2}(\omega)}$$

Magnitude-squared Coherence (MSC)

$$C_{xy}(\omega) = |\text{Coh}_{xy}(\omega)|^2 = \frac{|S_{xy}(\omega)|^2}{S_{xx}(\omega)\, S_{yy}(\omega)}$$

Relationship between signals

$C_{xy}(\omega) = 1 \Rightarrow x[n], y[n]$ related through LTI system

$C_{xy}(\omega) = 0 \Rightarrow x[n], y[n]$ uncorrelated

Estimation of cross-spectra follows estimation of power spectra

Periodogram inspired: $\hat{S}_{xy}(\omega) = \frac{1}{L} X(e^{j\omega}) Y^*(e^{j\omega})$

length L: $x[n] \overset{DTFT}{\longleftrightarrow} X(e^{j\omega})$, $y[n] \overset{DTFT}{\longleftrightarrow} Y(e^{j\omega})$

- suffers from high variance
- fails for MSC

$$\hat{C}_{xy}(\omega) = \frac{\frac{1}{L} X(e^{j\omega}) Y^*(e^{j\omega}) \; \frac{1}{L} X^*(e^{j\omega}) Y(e^{j\omega})}{\frac{1}{L} X(e^{j\omega}) X^*(e^{j\omega}) \; \frac{1}{L} Y(e^{j\omega}) Y^*(e^{j\omega})} = 1$$

Averaging approaches, e.g. Welch's method

- averaging reduces variance
- meaningful MSC

Welch's Method for MSC

3:39

- Divide $x[n], y[n]$ into length L segments
  Offset R (overlap $L-R$)
  $(K-1)R+L \leq Q$ (total data length

- Take DFT of each segment
  $x_k[n] \overset{DTFT}{\longleftrightarrow} X_k(e^{j\omega})$, $y_k[n] \overset{DTFT}{\longleftrightarrow} Y_k(e^{j\omega})$

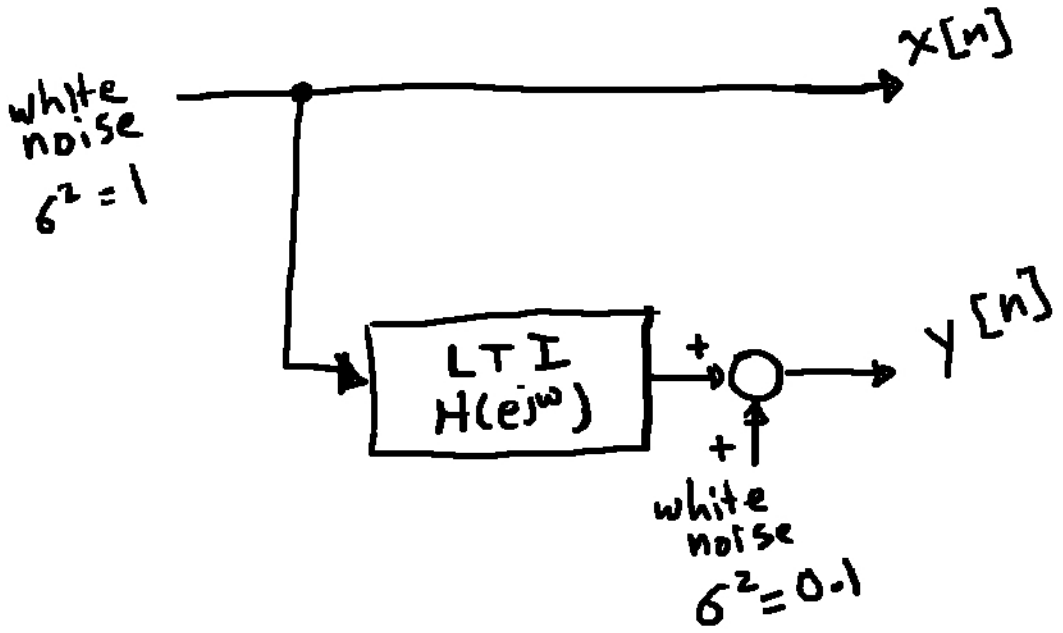
- Form power/cross spectra for each segment
  $I_{xx}^k(\omega) = \frac{1}{L}|X_k(e^{j\omega})|^2$, $I_{yy}^k(\omega) = \frac{1}{L}|Y_k(e^{j\omega})|^2$, $I_{xy}^k = \frac{1}{L}X(e^{j\omega})Y^*(e^{j\omega})$

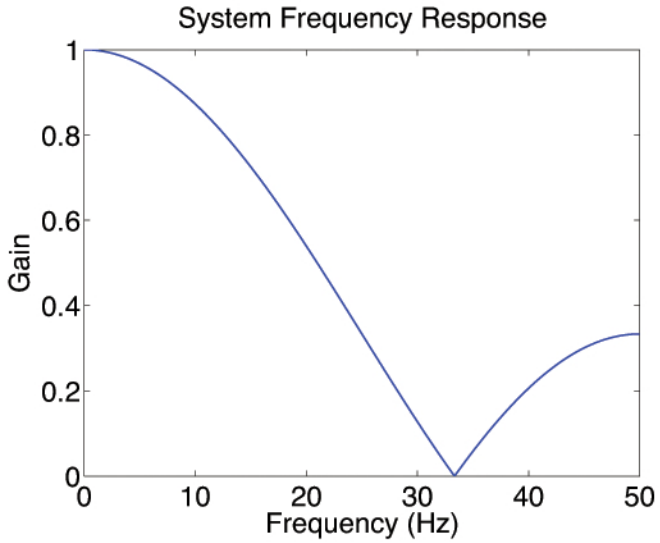
- Average power/cross spectra

$$\hat{C}_{xy}(\omega) = \frac{\left|\frac{1}{K}\sum_{k=1}^{K} I_{xy}^k(\omega)\right|^2}{\frac{1}{K}\sum_{k=1}^{K} I_{xx}^k(\omega)\ \frac{1}{K}\sum_{k=1}^{K} I_{yy}^k(\omega)}$$

Example: mscohere in MATLAB

5:22



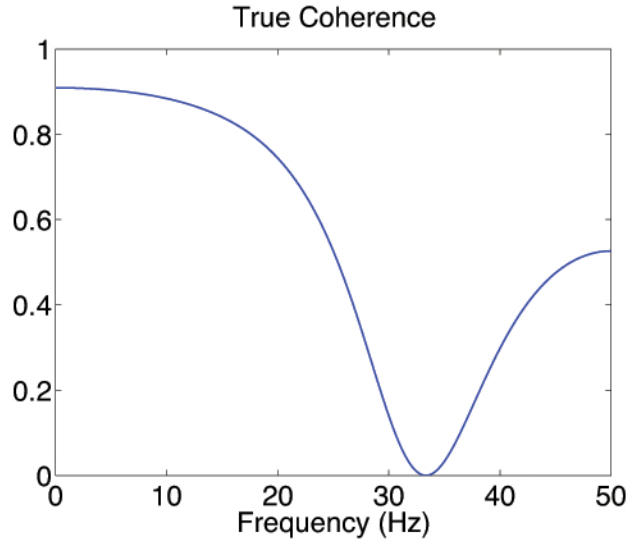


$$S_{xx}(\omega) = 1$$

$$S_{yy}(\omega) = |H(e^{j\omega})|^2 + 0.1$$

$$S_{xy}(\omega) = |H(e^{j\omega})|^2$$

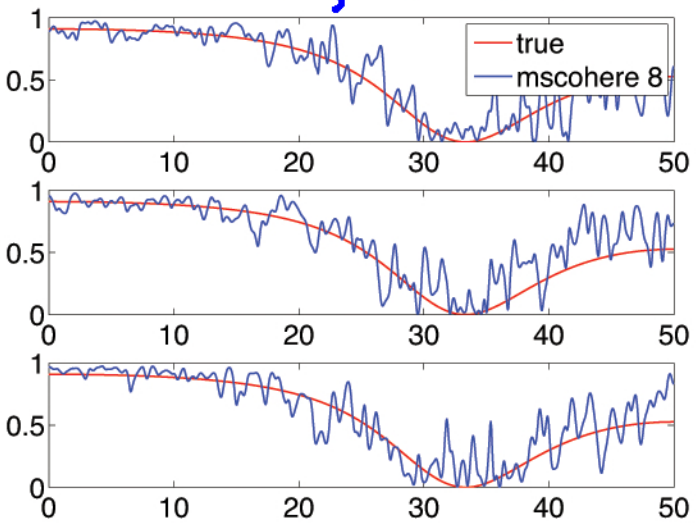
$$C_{xy}(\omega) = \frac{|H(e^{j\omega})|^2}{|H(e^{j\omega})|^2 + 0.1}$$

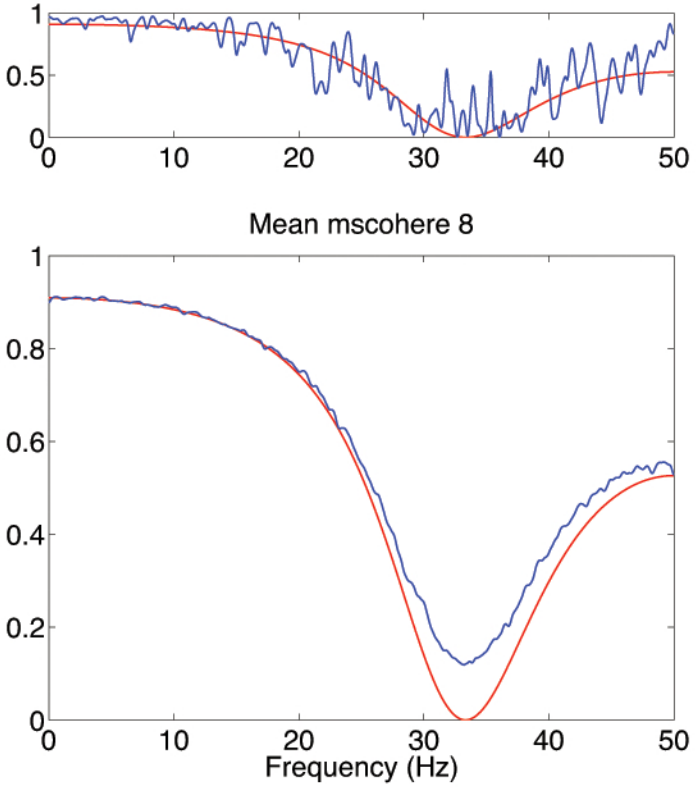


8:16

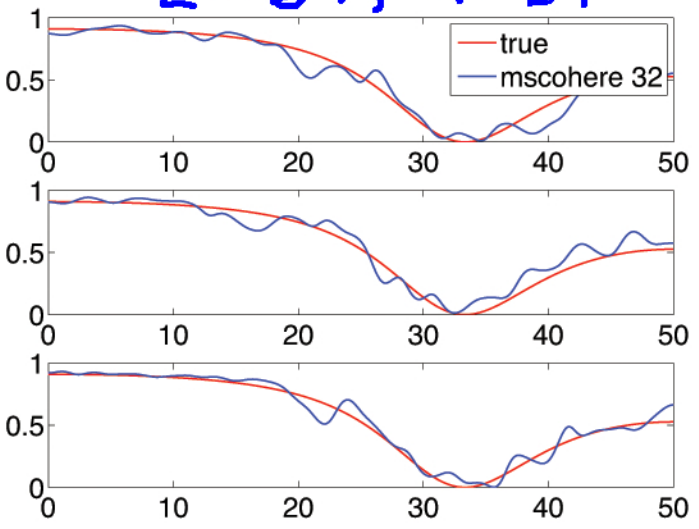
Q = 2048

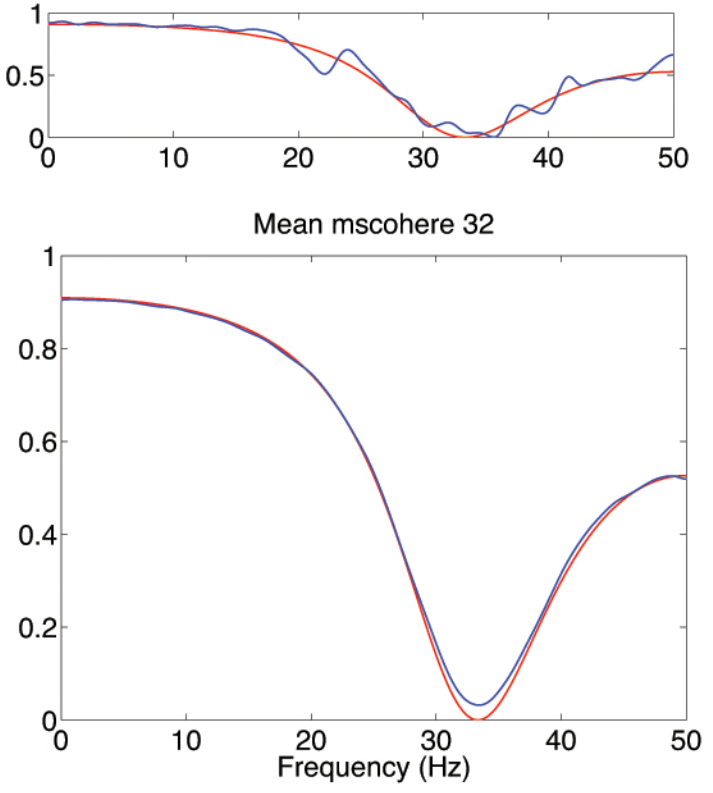
K = 8
L = 256, R = 256



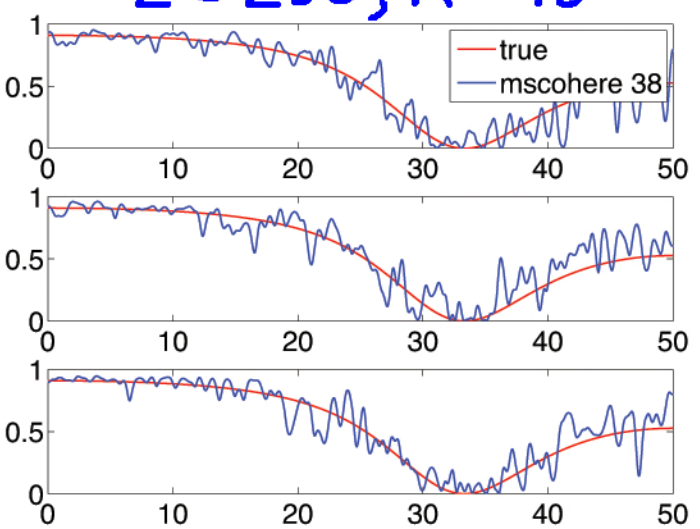


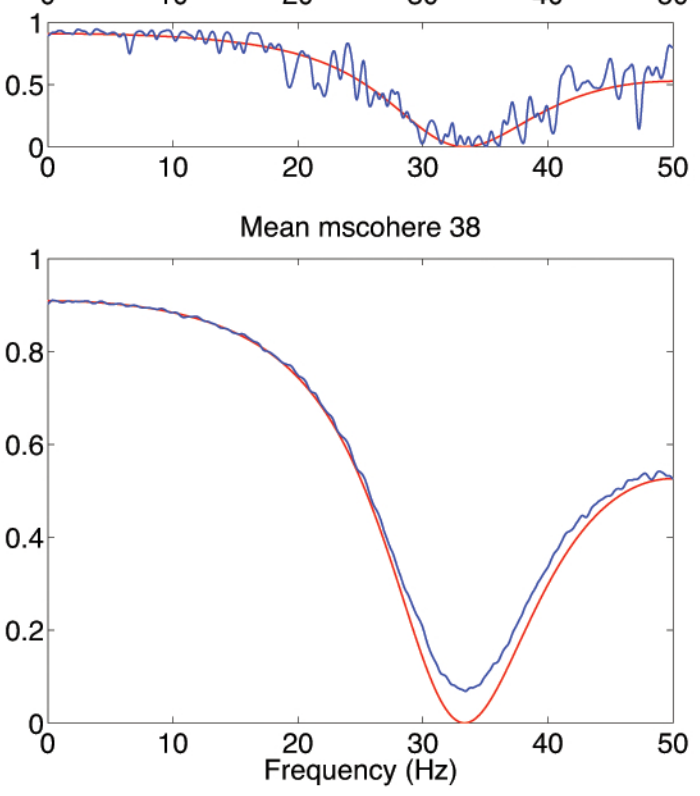
K = 32
L = 64, R = 64





K = 38
L = 256, R = 48





For more great educational signal-processing content:

AllSignalProcessing.com

- Intuitive, easy-to-follow style
- Organization that saves you time
- Answers to your questions
- Learn at your pace
- Award winning instruction